

Automated Progression Monitoring of Pulmonary Nodules across Follow-up Lung CT scans

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Introduction

- Automated tracking of pulmonary nodules and their progression monitoring plays an important role in the early detection of lung cancer.
- Here we demonstrated our automated detection and tracking tool performance on follow up scans with the ground truth labels annotated by the radiologist



Dataset Annotations

	First visit	Second visit	Third visit
27 Patient	63 nodules	63 nodules	63 nodules
87 Patient	219 nodules	219 nodules	

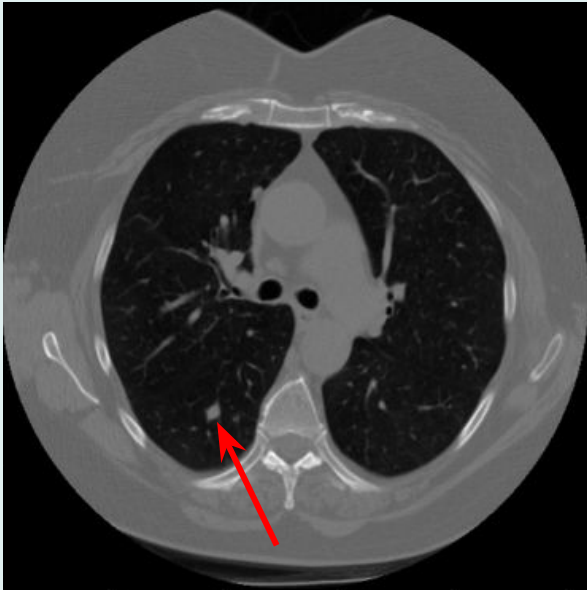
Radiologist with 5-year experience annotated(nodule mask) the same nodules present at the baseline (first visit of a patient) in next follow-ups (second and third visit)

Total: 282 nodules present at the baseline

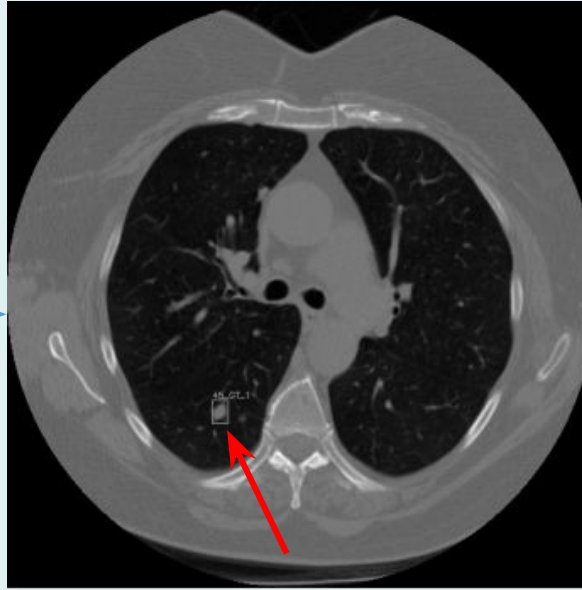


Methods

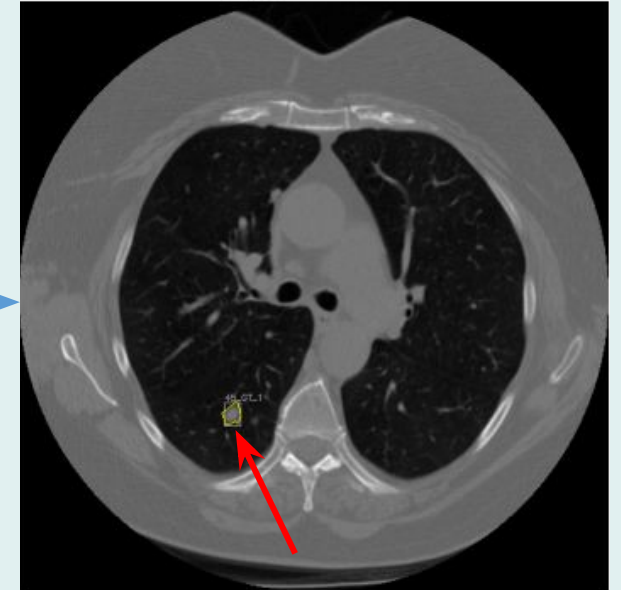
Pulmonary Nodules Detection and Segmentation



Given a chest CT scan
having pulmonary
nodules

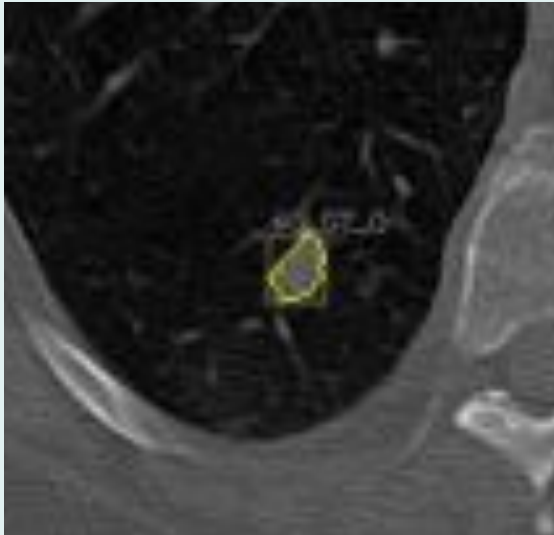


Deep learning based nodule
detection model identify the
pulmonary nodules and output a
bounding box around them



Deep learning based
segmentation model is used to
segment the nodules inside the
predicted bounding box from the
detection model

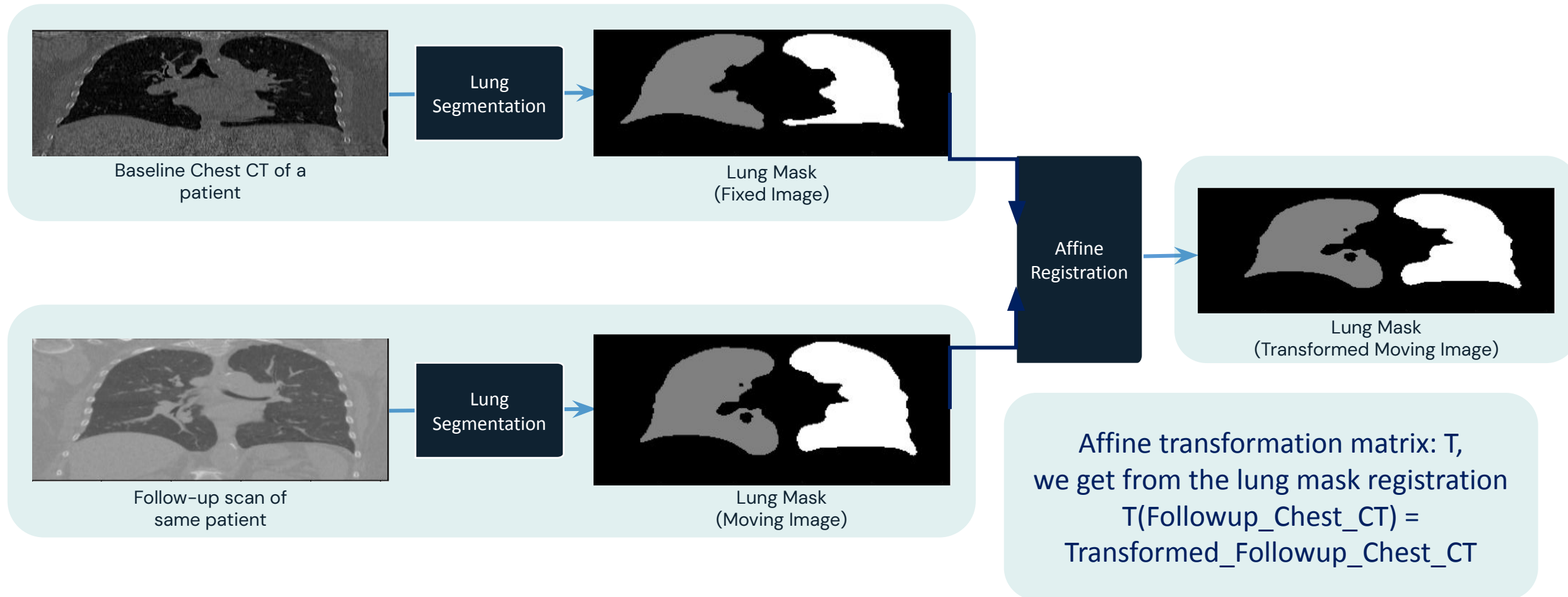
Volume Calculation



Segmentation mask of nodules and spacing of chest CT scan is needed to calculate volume

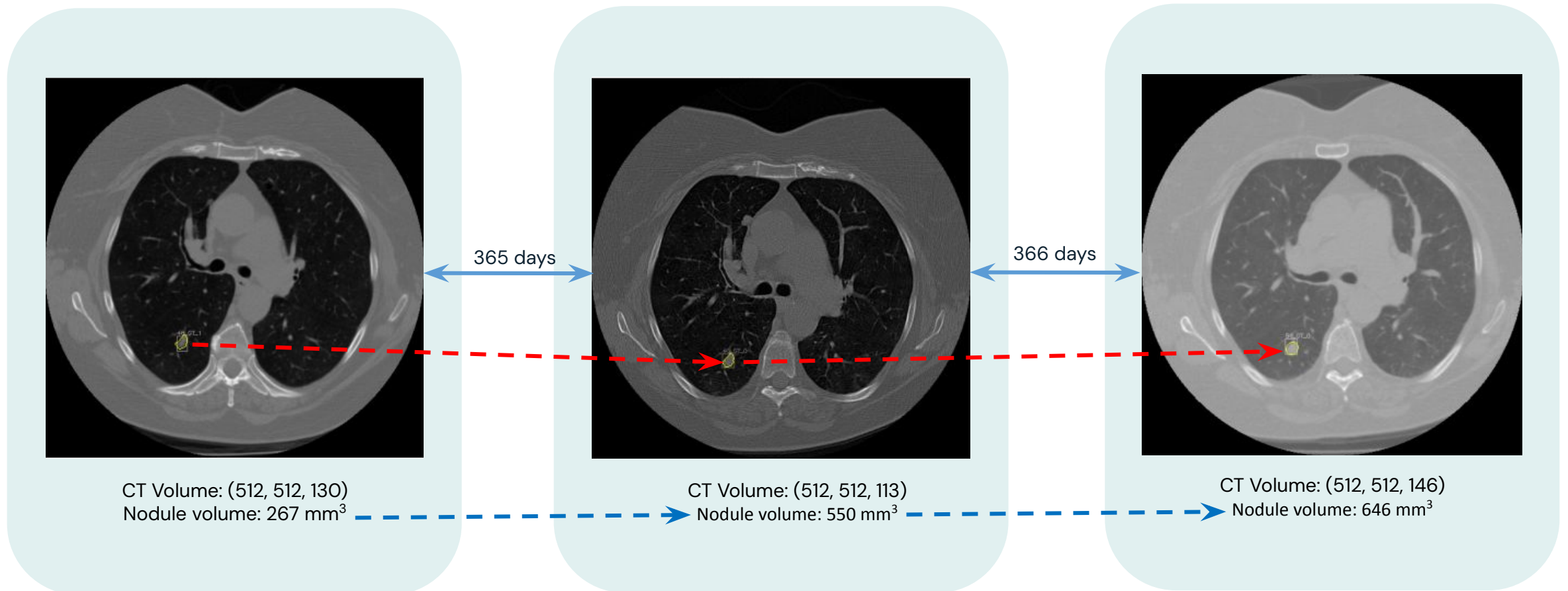
Nodule volume = (No. of voxels present in segmentation mask)*(CT spacing)

Pulmonary Nodules Tracking Algorithm



Using registration discussed in previous slide, follow-up chest CT are aligned to baseline chest CT,
After alignment, same nodules at the baseline can be found at the same location of the follow-up scan.

Pulmonary Nodules Tracking Across Follow-ups



Results: Nodule Tracking

	First visit	Second visit	Third visit
27 Patient	63 nodules	52/63 nodules	52/63 nodules
87 Patient	219 nodules	207/219 nodules	

Our Algorithm correctly track 52 of 63 nodules in both follow-ups and 207 nodules out of 219 nodules in first follow-up.

Total: 282 nodules present at the baseline

Results: Progression Monitoring

- VDT calculation:
 - $VDT = [\ln_2 \times \Delta T] / [\ln(X_2 / X_1)]$
 - X_2 and X_1 = the final and initial nodule volumes
 - ΔT = time (in days) between the two scans
 - $\ln$ = natural logarithm

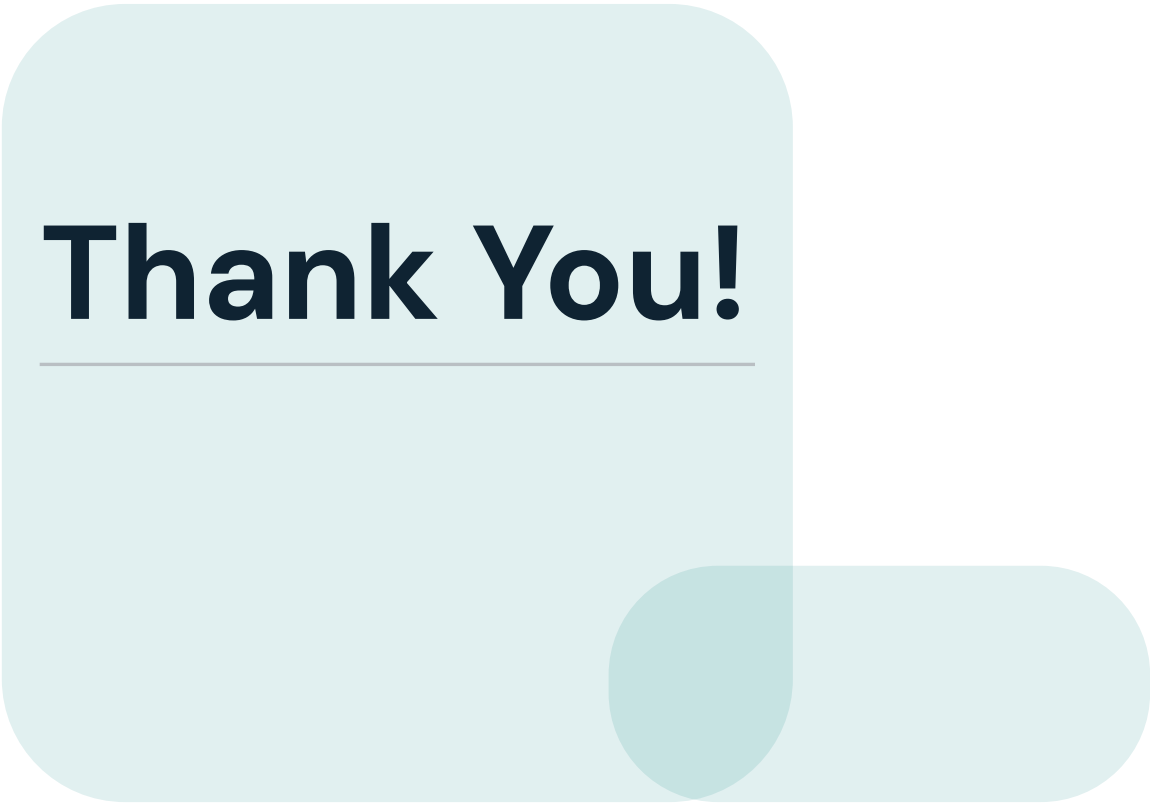
In ground truth, there are 76 nodules which have $VDT < 400$ days,
Nodule Segmentation model correctly predict segmentation mask leads to correct volume determination,
We able to track 54 nodules out of 76 nodules having $VDT < 400$ days.

Conclusion

- **Our algorithm can automate nodule tracking leads to automate the progression monitoring of nodules which may become malignant in follow-ups, thereby increase efficiency and improve patient management**

References

1. MacMahon, H., Naidich, D.P., Goo, J.M., Lee, K.S., Leung, A.N., Mayo, J.R., Mehta, A.C., Ohno, Y., Powell, C.A., Prokop, M. and Rubin, G.D., 2017. Guidelines for management of incidental pulmonary nodules detected on CT images: from the Fleischner Society 2017. *Radiology*, 284(1), pp.228–243.
2. Devaraj, A., van Ginneken, B., Nair, A. and Baldwin, D., 2017. Use of volumetry for lung nodule management: theory and practice. *Radiology*, 284(3), pp.630–644.

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Thank You!
